

YEAR 11
ASSESSMENT TASK 3
2019

MATHEMATICS
EXTENSION 1

General Instructions	• Reading time – 5 minutes • Working time – 2 hours • Write using black pen • Calculators approved by NESA may be used • A separate reference sheet is provided • For questions in Section II, show relevant mathematical reasoning and/or calculations
Total marks: 85	Section I – 5 marks • Attempt Questions 1–5 • Allow about 7 minutes for this section Section II – 80 marks • Attempt Questions 6–9 • Allow about 1 hours and 53 minutes for this section

The answers to all questions are to be returned in separate *stapled* bundles, clearly marked with Question 6, Question 7, etc., with your student number.

Question 1

Which set provides a solution to the inequality $|3x - 2| \geq |x + 4|$?

- A. $\{x: -\frac{1}{2} \leq x \leq 3\}$
- B. $\{x: x \leq -\frac{1}{2} \text{ or } x \geq 3\}$
- C. $\{x: x \leq -\frac{1}{2}\}$
- D. $\{x: x \leq 3\}$

Question 2

Let $P(x) = x^3 + 3x^2 + ax + b$, where a and b are integers. When $P(x)$ is divided by $(x + 1)$, the remainder is 4. When $P(x)$ is divided by $(x - 1)$, the remainder is -2 .

What is the remainder when $P(x)$ is divided by $(x^2 - 1)$?

- A. $4x + 2$
- B. $-4x - 2$
- C. $3x - 1$
- D. $-3x + 1$

Question 3

The cubic equation $x^3 + px^2 + qx + r = 0$, where p, q and r are integers, has roots α, β and γ , such that $\alpha + \beta + \gamma = 15$ and $\alpha^2 + \beta^2 + \gamma^2 = 83$.

What is the value of $p + q$?

- A. 56
- B. -56
- C. -86
- D. 86

Question 4

A team of six students is to be formed from a class of ten students. How many different teams can be formed if two particular students cannot both be selected for the team?

- A. 252
- B. 210
- C. 168
- D. 140

Question 5

What does the expression $\frac{\cos 3x - \cos 5x}{\sin 3x + \sin 5x}$ simplify to?

- A. $\cot x$
- B. $-\tan x$
- C. $\tan x$
- D. $-\cot x$

Question 6 (20 Marks)**START A NEW PAGE****Marks**

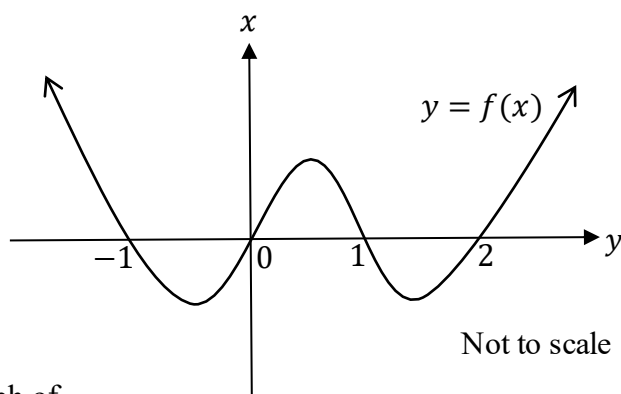
- (a) Find the exact value of: $\sin\left(\sin^{-1}\frac{1}{2} + \cos^{-1}\frac{5}{13}\right)$ **2**
- (b) If $\frac{\sin \theta}{1 + \cos \theta} + \frac{\sin \theta}{1 - \cos \theta} = 4$, find the exact value of $\cot 2\theta$ for $0 \leq \theta \leq \frac{\pi}{2}$. **4**
- (c) Show that $\cos 37\frac{1}{2}^\circ \sin 7\frac{1}{2}^\circ = \frac{\sqrt{2}-1}{4}$, without using a calculator. **2**
- (d) Sketch the graph of $y = \frac{1}{2}\cos^{-1} 2x$. **2**
- (e) Consider the function $f(x) = \frac{x-1}{x-2}$
- (i) Show that $f^{-1}(x) = \frac{2x-1}{x-1}$ **1**
- (ii) Sketch f and f^{-1} on the same system of axes, showing any asymptotes and intercepts. **3**
Clearly label your graphs.
- (f) The speed, $V \text{ m s}^{-1}$, of a parachute, t seconds after jumping from an aeroplane is modelled by the equation
- $$V = 42\left(1 - \frac{1}{e^{\pi t}}\right).$$
- (i) Find the acceleration of the parachutist after π seconds, to two decimal places. **2**
- (ii) The parachute opens when it reaches a speed of 21 m s^{-1} . **2**
Find the exact time of falling before the parachute opens?
- (g) Show that among the 900 students at James Ruse, at least 3 students share a birthday. **2**

(a) For what values of x is the inequality $\frac{x}{x+1} \geq \frac{2}{x+3}$ satisfied? 4

(b) Find the values of a and b if $2x^3 - (2a+1)x^2 + (2+b)x - 1 = 0$ has a multiple root at $x = 1$. 3

(c) The parametric equations of a curve are $x = \ln(1+t^2)$ and $y+1 = \ln(1+2t^2)$. 4
Find the cartesian equation of the curve, and hence show that the x -intercept of the curve is $\ln\left(\frac{1+e}{2}\right)$.

(d) The graph of $y = f(x)$ is shown.



On separate systems of axes, draw the graph of

(i) $y = f(|x|)$ 2

(ii) $y = \sqrt{(f(x))^2}$ 2

(e) From seven girls and five boys, a committee of seven is to be chosen. 2
What is the probability of choosing a committee containing at least four girls?

(f) How many people would have to be in a school before it contained at least two people 3
with the same first and last initials.

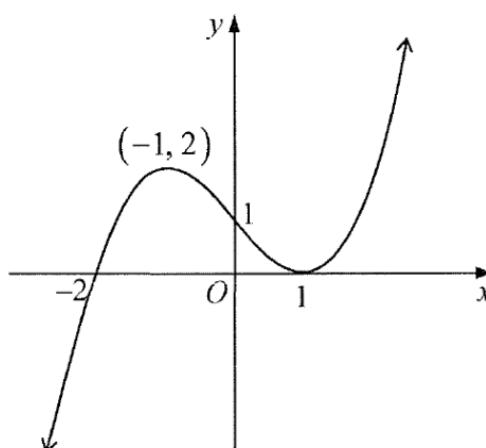
- (a) Solve for x and y : 3

$$2 \tan^{-1} x - \cos^{-1} y = \frac{\pi}{2}$$

$$3 \cos^{-1} y + \tan^{-1} x = \frac{5\pi}{6}$$

- (b) Solve $\cos x - \cos 3x = 0$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. 3

- (c) The graph of $y = f(x)$ is drawn below.



Draw a separate half-page graph for each of the following functions, showing all important features, asymptotes and intercepts.

- (i) $y^2 = f(x)$ 3

- (ii) $y = \frac{1}{f(x)}$ 3

- (d) At a particular dinner, each rectangular table has nine seats, five facing the stage and four with their backs to the stage.

- (i) In how many ways can 9 people be seated at the table if John and Mary sit on the same side? 2

- (ii) What is the probability of John and Mary sitting on opposite sides of the table? 3

- (e) 10 points are placed randomly in a 1 by 1 square. Show that there must be some pair of points that are within $\frac{\sqrt{2}}{3}$ of each other. 3

- (a) (i) If $t = \tan \theta$, show that $\tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}$ 3
- (ii) Given the roots of $\tan 4\theta = \cot \theta$ are $\theta = \frac{\pi}{10}$ and $\theta = \frac{3\pi}{10}$. 3
Find the exact value of $\tan \frac{\pi}{10}$.
- (b) If α, β and γ are the roots of $3x^3 + 8x^2 - 1 = 0$, find the value of 3
 $\left(\beta + \frac{1}{\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right)\left(\alpha + \frac{1}{\beta}\right)$.
- (c) A metal rod is taken from a freezer at -8°C into a room where the air temperature is 22°C . The rate at which the rod warms follows Newton's law, that is $\frac{dT}{dt} = -k(T - 22)$ where k is a positive integer, time t is measured in minutes and temperature T in $^\circ\text{C}$.
- (i) Show that the function $T = 22 - Ae^{-kt}$, where A is a constant, provides this rate of change. 1
- (ii) Hence find the value of A 2
- (iii) The temperature of the rod reaches 4°C in 90 minutes. 2
Find the exact value of k .
- (iv) Find the temperature of the rod after another 90 minutes. 1
- (d) Suppose a particular population of bacteria obeys the growth formula $P(t) = \frac{6000}{3 + 7e^{-0.2t}}$ where P is measured in milligrams and time t , in hours.
- (i) Predict what the population will be as t gets very large. 1
- (ii) If the population grows the fastest when $P(t) = 1000$, find when this occurs, to 4 significant figures. 2
- (c) Sketch the graph of $P(t)$, showing all important features. 2

END OF EXAMINATION

Question	6		7		8		9		Total
Functions		/4		/18		/6		/3	/31
Combinatorics		/2		/2		/8		-	/12
Trigonometric Functions		/10		-		/6		/6	/22
Calculus		/4		-		-		/11	/15
MCQ									/5
Total	/5	/20		/20		/20		/20	/85

Justification for Q4: Students: X X X X X X X A B

If neither A or B: 28 possibilities (8C6 from non A and B)

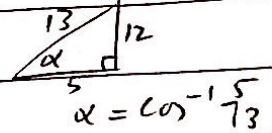
If A or B: 112 possibilities (8C5 x 2)

Question 6

$$\begin{aligned}
 a) \quad & \sin \left(\sin^{-1} \frac{1}{2} + \cos^{-1} \frac{5}{13} \right) \\
 &= \sin \left(\frac{\pi}{6} + \cos^{-1} \frac{5}{13} \right) \\
 &= \sin \frac{\pi}{6} \cdot \cos \left(\cos^{-1} \frac{5}{13} \right) + \sin \left(\cos^{-1} \frac{5}{13} \right) \cdot \cos \frac{\pi}{6}
 \end{aligned}$$

$$1 \text{ m} \quad = \quad \frac{1}{2} \cdot \frac{5}{13} + \frac{12}{13} \cdot \frac{\sqrt{3}}{2}$$

$$1 \text{ m} \quad = \quad \frac{5 + 12\sqrt{3}}{26}$$



$$b) \quad \frac{\sin \theta}{1 + \cos \theta} + \frac{\sin \theta}{1 - \cos \theta} = 4$$

$$\frac{\sin \theta - \sin \theta \cos \theta + \sin \theta + \sin \theta \cos \theta}{1 - \cos^2 \theta} = 4$$

1 m domain restriction

$$\cos \theta \neq 1 \Rightarrow \theta \neq 0$$

(or similar)

$$\frac{2 \sin \theta}{\sin^2 \theta} = 4$$

$$2 \sin^2 \theta - \sin \theta = 0$$

$$\sin \theta (2 \sin \theta - 1) = 0$$

$$1 \text{ m} \quad \sin \theta = 0 \text{ or } \sin \theta = \frac{1}{2} \quad \left(\text{for domain, } \theta = 0^\circ, \frac{\pi}{6} \right)$$

$$\therefore \cot 2\theta = \frac{1}{\tan 2\theta} = \frac{1 - \tan^2 \theta}{2 \tan \theta}$$

$$1 \text{ m} \quad = \frac{1 - \left(\frac{1}{\sqrt{3}}\right)^2}{2/\sqrt{3}}$$

$$1 \text{ m} \quad = \frac{\sqrt{3}}{3} \quad \text{only as } \cot 2\theta \text{ undefined for } \theta = 0$$



(c) $\cos 37\frac{1}{2}^\circ \cdot \sin 7\frac{1}{2}^\circ$

1 m $= \frac{1}{2} [\sin 45^\circ - \sin 30^\circ]$

$= \frac{1}{2} \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right)$

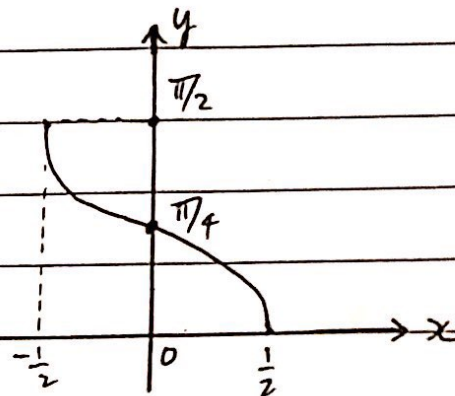
$= \frac{1}{2} \left(\frac{\sqrt{2}-1}{2} \right)$

1 m $= \frac{1}{4} (\sqrt{2}-1)$

d) $2y = \cos^{-1} 2x$

1 m $\left\{ \begin{array}{l} \text{Range: } 0 \leq 2y \leq \pi \\ 0 \leq y \leq \frac{\pi}{2} \end{array} \right.$

$\left\{ \begin{array}{l} \text{Domain: } -1 \leq 2x \leq 1 \\ -\frac{1}{2} \leq x \leq \frac{1}{2} \end{array} \right.$



1 m for shape

Note vertical tangents at $x = \pm \frac{1}{2}$

e) (i) $x = \frac{y-1}{y-2}$

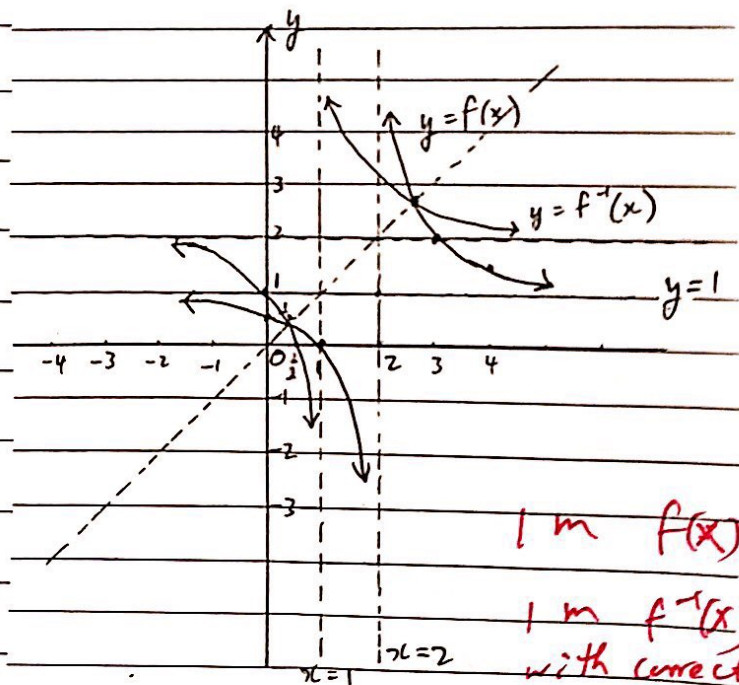
(ii)

$xy - 2x - y + 1 = 0$

$y(x-1) = 2x-1$

1 m $\therefore y = \frac{2x-1}{x-1}$

$\therefore f^{-1}(x) = \frac{2x-1}{x-1}$



1 m $f(x)$

1 m $f^{-1}(x)$

with correct intercepts &

asymptotes

1 m symmetrical about $y=x$



f) $V = 42(1 - e^{-t/\pi})$

1 m (i) $a = \frac{dv}{dt} = \frac{42}{\pi} e^{-t/\pi}$

v. poorly done

$$\therefore a(\pi) = \frac{42}{\pi} e^{-1}$$

$$= \frac{42}{\pi e}$$

$$= 4.9181\dots$$

1 m

$$= 4.92 \text{ m/s}^2$$

Need to be to 2 d.p.

1 m (ii) $21 = 42(1 - e^{-t/\pi})$

$$-e^{-t/\pi} = -\frac{1}{2}$$

$$\therefore t/\pi = \ln 2$$

1 m

$$\therefore t = \pi \ln 2 \text{ sec.}$$

Some students
continue to give
decimal places
won't get the second
mark



1. (g)

If there are 366 students and 366 days, then no student may share a birthday.

However with 1 more, at least 2 students may.

Hence if $2 \times 366 = 732$ students at least 2 may.

However with 1 more than 732, at least 3 may.

So with 900 students at least 3 may share a birthday, by the PHP.

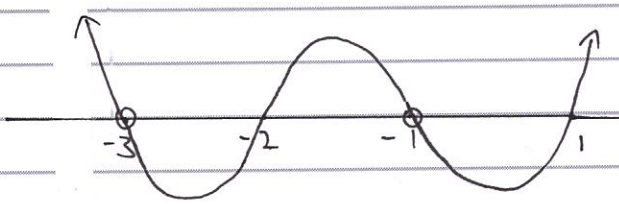
[Similar argument if 365 days in a year are considered.]

OR From $n = dq + r$,

$$900 = 2 \times 366 + 168$$

By PHP, at least $(2+1)$ students may share a birthday.

MATHEMATICS Extension 1 : Question...7....

Suggested Solutions	Marks	Marker's Comments
a) $\frac{x}{x+1} \geq \frac{2}{x+3}$ $x(x+1)(x+3)^2 \geq 2(x+3)(x+1)^2$ $x(x+1)(x+3)^2 - 2(x+3)(x+1)^2 \geq 0$ $(x+1)(x+3)(x^2+3x-2x-2) \geq 0$ $(x+1)(x+3)(x^2+x-2) \geq 0$ $(x+1)(x+3)(x+2)(x-1) \geq 0$  Since $x \neq -3$ or -1 $x < -3$ or $-2 \leq x < -1$ or $x \geq 1$ <u>or</u> $\frac{x}{x+1} \geq \frac{2}{x+3}$ $\frac{x}{x+1} - \frac{2}{x+3} \geq 0$ $\frac{x^2+3x-2x-2}{(x+1)(x+3)} \geq 0$ $\frac{x^2+x-2}{(x+1)(x+3)} \geq 0$ $\frac{(x+2)(x-1)}{(x+1)(x+3)} \geq 0$ Since LHS has the same sign as $(x+3)(x+2)(x+1)(x-1)$ then		① ① ① Inequality without the restrictions ② correct answer with restrictions ① ① draw polynomials like before

Anyone who multiplied both sides by terms that are not necessarily positive to begin with, will receive 2 maximum provided they did everything else correctly.

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
a) Continued. Don't square both sides !!! When solving $a^2 \geq b^2$ you are solving for $a > b$ (when a is positive) <u>AND</u> $a < b$ (when a is negative) b) let $f(x) = 2x^3 - (2a+1)x^2 + (2+b)x - 1$ $f'(x) = 6x^2 - (4a+2)x + (2+b)$ $f'(1) = 0 \Rightarrow 6(1)^2 - (4a+2)(1) + (2+b) = 0$ $6 - 4a - 2 + 2 + b = 0$ $4a - b = 6$ - Egn 1 $f(1) = 0 \Rightarrow 2(1)^3 - (2a+1)(1)^2 + (2+b)(1) - 1 = 0$ $2 - 2a - 1 + 2 + b - 1 = 0$ $2a - b = 2$ - Egn 2 Egn 1 - Egn 2 $\rightarrow 2a = 4$ $a = 2$ Sub $a = 2$ into Egn 1. $2(2) - b = 2$ $4 - b = 2$ $b = 2$ $\therefore a = 2$ $b = 2$		① ① If Someone did the exact same steps but made an algebraic error in each of the 3 steps, they still received 1/3. ①

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
$c) \quad x = \ln(1+t^2) \quad y+1 = \ln(1+2t^2)$		
$1+t^2 = e^x \quad 1+2t^2 = e^{y+1}$		① Raising e to the power of both sides
$t^2 = e^x - 1 \quad 2t^2 = e^{y+1} - 1$		①
$t^2 = \frac{e^{y+1} - 1}{2}$		Make t^2 the subject for both equations
$\therefore e^x - 1 = \frac{e^{y+1} - 1}{2}$		
$2e^x - 2 = e^{y+1} - 1$		
$2e^x - e^{y+1} - 1 = 0$		① Simplifying the constants from both sides
When $y=0$		
$2e^x - e - 1 = 0$		
$2e^x = e + 1$		
$e^x = \frac{e+1}{2}$		
$x = \ln\left(\frac{e+1}{2}\right)$		① for getting to the result without skipping steps
<u>Note</u>: Cartesian equation can be simplified to		
$y = \ln(2e^x - 1) - 1$		
but not necessary.		

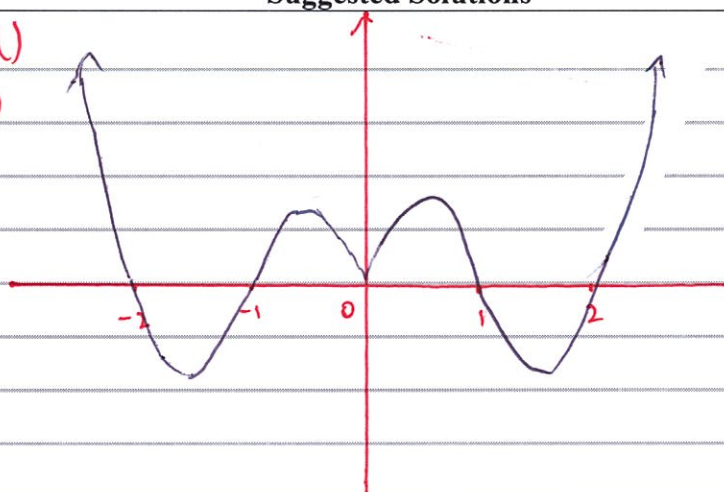
MATHEMATICS Extension 1 : Question.....

Suggested Solutions

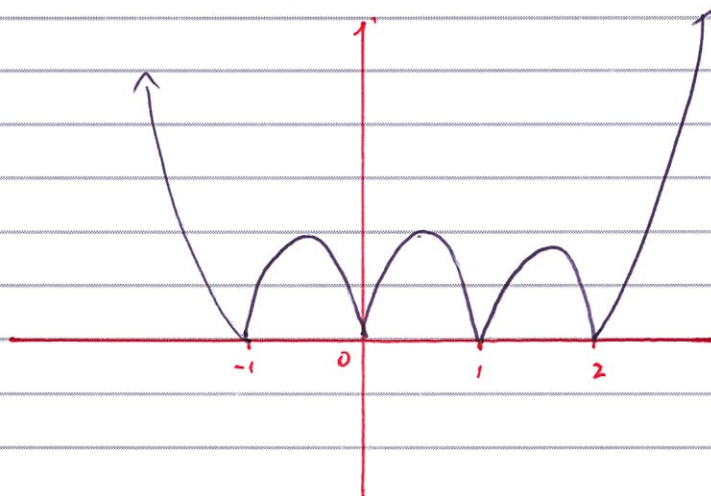
Marks

Marker's Comments

d)
i)



ii)



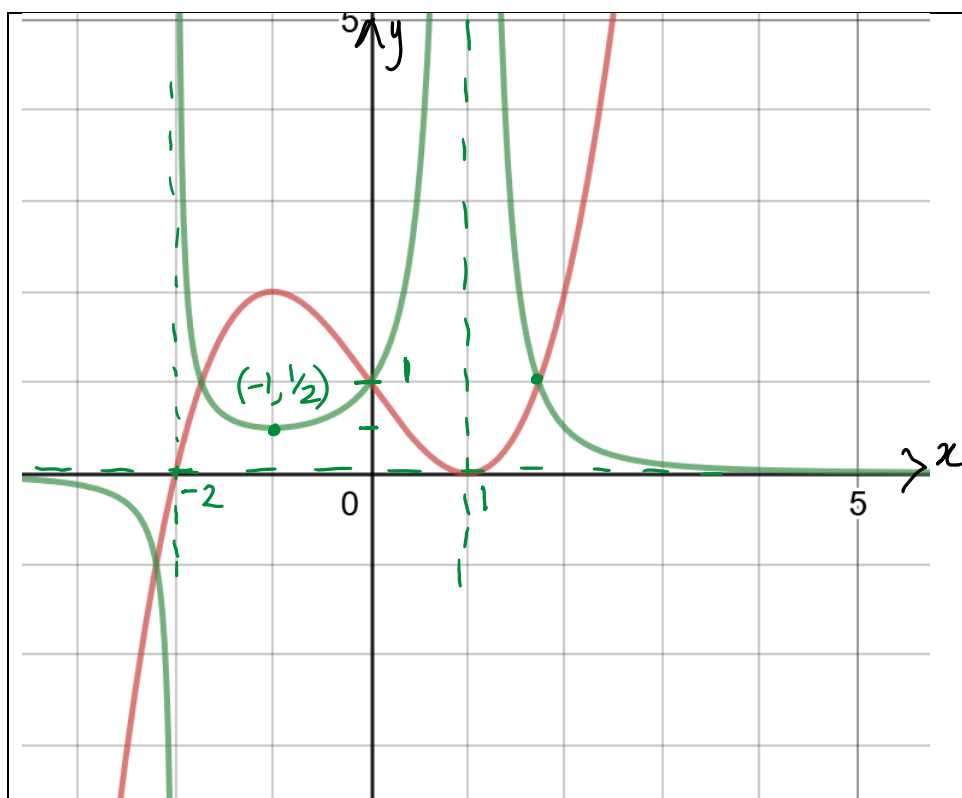
① Mark subtracted per mistake

MATHEMATICS Extension 1 : Question.....

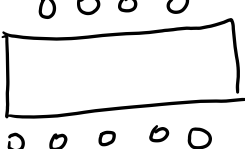
Suggested Solutions	Marks	Marker's Comments
e) Total = $N(4 \text{ girls}) + N(5 \text{ girls}) + N(6 \text{ girls}) + N(7 \text{ girls})$ $= ({}^7C_4 \times {}^5C_3) + ({}^7C_5 \times {}^5C_2) + ({}^7C_6 \times {}^5C_1) + ({}^7C_7 \times {}^5C_0)$ $= 350 + 210 + 35 + 1$ $= 596$ $N(\text{no restrictions}) = {}^{12}C_7 = 792$ $\therefore \text{Probability}(\geq 4 \text{ girls}) = \frac{596}{792} = \frac{149}{198}$		*Students also received 1 mark for $P = (4 \text{ girls}) + (5 \text{ girls}) + (6 \text{ girls}) + (7 \text{ girls})$ $\frac{\quad}{{}^{12}C_7}$ ①
f) There are 26 letters in the alphabet $\therefore$ There are 26^2 different initials combinations $\therefore$ Number of students required $= 26^2 + 1$ $= 676 + 1$ $= 677$		① ①

MATHEMATICS Extension 1: Question 8

Suggested Solutions	Marks	Marker's Comments
a) $2\tan^{-1}x - \cos^{-1}y = \pi/2$ ① $2\tan^{-1}x + 6\cos^{-1}y = 5\pi/3$ ② ② - ① $7\cos^{-1}y = \pi\pi/6$ $\cos^{-1}y = \pi/6$ $\therefore y = \sqrt{3}/2$ Sub into ①: $2\tan^{-1}x = \pi/2 + \pi/6$ $\tan^{-1}x = \pi/3$ $\therefore x = \sqrt{3}$	1 1 1	eliminating x exact value of y exact value of x
b) $\cos x - \cos 3x = 0$ $\cos(2x-x) - \cos(2x+x) = 0$ $2\sin 2x \cdot \sin x = 0$ $\therefore \sin 2x = 0$ or $\sin x = 0$ $2x = 0, \pi, -\pi$ or $x = 0$ $\therefore x = 0$ or $\pm \pi/2$	1 1 1 1	identity $x = 0$ $x = \pm \pi/2$
c) (i)	1 1 1	symmetry about x-axis $(-1, \pm\sqrt{2})$ intercepts at $x = 1, -2$ $y = 1, -1$ shape particularly at $x = 1$ and as $x \rightarrow \infty$



- 1 asymptotes and y-intercept of 1
 - 1 $(-1, \frac{1}{2})$ as a minimum
 - 1 shape and axes labelled
- mostly well done.

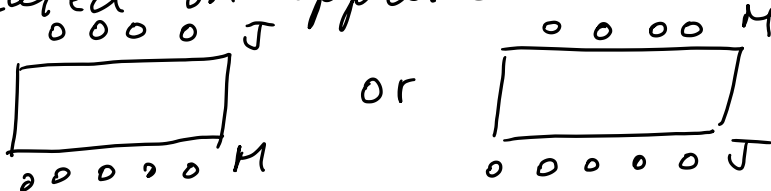
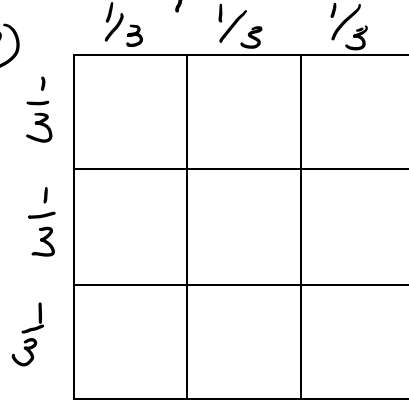
d)  facing stage:
 J has 5 to choose from
 M has 4 to choose from
 backs to stage:
 J has 4 to choose from
 M has 3 to choose from
 other 7 arranged in $7!$ ways
 $\therefore$ J & M can be seated in $4 \times 3 \times 7! + 5 \times 4 \times 7!$
 $= 7! (12 + 20)$
 $= 161280$ ways

- 1 no. of ways facing stage
- 1 no. of ways back to stage

(ii) Method 1:
 Total no. of arrangements = $9!$
 $P(\text{J \& M on opposite sides})$
 $= 1 - P(\text{J \& M on same side})$
 $= 1 - \frac{161280}{9!}$
 $= \frac{5}{9}$

- 1
- 1
- 1

MATHEMATICS Extension 1: Question 8

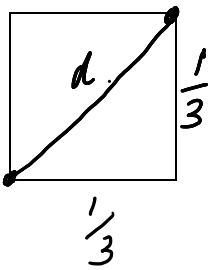
Suggested Solutions	Marks	Marker's Comments
Method 2: NO. of ways J & M can be seated on opposite sides:  or No. of ways = $4 \times 5 \times 7! + 4 \times 5 \times 7!$ $= 40 \times 7!$ $= 201600$ $P(J \& M \text{ on opposite sides})$ $= \frac{201600}{9!}$ $= \frac{5}{9}$ e)  Divide the 1×1 square into 9 smaller squares of size $\frac{1}{3} \times \frac{1}{3}$. Let the 10 points be the pigeons and the 9 smaller squares be the pigeonholes. Worst case scenario, 9 points are placed randomly inside the 1×1 square so that there is one point in each smaller square. When a 10th point is placed inside the square there must now be	1 1 1 1 1	total arrangements explanation of dividing 1×1 square into smaller squares of size $\frac{1}{3} \times \frac{1}{3}$ units use of Pigeon hole principle: 10 pigeons into 9 pigeonholes

Suggested Solutions

Marks

Marker's Comments

one smaller square which contains 2 points. The furthest these 2 pts can be apart is when they are placed at either end of the diagonal in a $\frac{1}{3} \times \frac{1}{3}$ square



$$d^2 = \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 \text{ by Pythagoras' Theorem}$$

$$= \frac{2}{9}$$

$$d = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3}$$

$\therefore$ There will be 2 points within $\frac{\sqrt{2}}{3}$ of each other

1

Pythagoras' Theorem with explanation and conclusion

Notes: Many students were not consistent in the language used, interchanging between squares/spaces/boxes/minisquare. Consistent language is necessary. Many students did not account for the points being placed randomly inside the 1×1 square and arranged the points around the square and then many fudged their answer. Many explanations need to be worded better.

MATHEMATICS Extension 1 : Question....9...

Suggested Solutions

Marks

Marker's Comments

a) i) RTP: $\tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}$

LHS = $\tan 4\theta$

= $\frac{2 \tan 2\theta}{1 - \tan^2 2\theta}$

= $2 \times \frac{2t}{1-t^2}$

= $\frac{4t}{1 - \left(\frac{2t}{1-t^2}\right)^2}$

= $\frac{4t}{1-t^2}$

= $\frac{4t}{1-t^2}$

= $\frac{4t}{1-t^2}$

= $\frac{4t}{1-t^2}$

= $\frac{4t(1-t^2)^2}{(1-t^2)^2 - 4t^2}$

= $\frac{4t(1-t^2)^2}{(1-t^2)^2 - 4t^2}$

= $\frac{4t(1-t^2)^2}{(1-t^2)^2 - 4t^2}$

= $\frac{4t(1-t^2)}{1-2t^2+t^4-4t^2}$

= $\frac{4t(1-t^2)}{t^4-6t^2+1}$

= $\frac{4t(1-t^2)}{t^4-6t^2+1}$

ii) $\tan 4\theta = \cot \theta$

$\frac{4t(1-t^2)}{t^4-6t^2+1} = \frac{1}{t}$

$4t^2(1-t^2) = t^4-6t^2+1$

$4t^2-4t^4 = t^4-6t^2+1$

$5t^4-10t^2+1=0$

$t^2 = \frac{10 \pm \sqrt{100-4(5)}}{10}$

= $\frac{10 \pm \sqrt{80}}{10}$

= $\frac{10 \pm \sqrt{80}}{10}$

1

⇒ shown some working towards final step.

1

MATHEMATICS Extension 1 : Question.....9

Suggested Solutions

Marks

Marker's Comments

$$\therefore \tan^2 \theta = \frac{5 \pm 2\sqrt{5}}{5}$$

$$\tan \theta = \pm \sqrt{\frac{5 \pm 2\sqrt{5}}{5}}$$

since $\theta = \frac{\pi}{10}$ is a solution,

$\tan \frac{\pi}{10}$ is in the 1st quadrant

$$\therefore \tan \frac{\pi}{10} > 0$$

$$\therefore \tan \frac{\pi}{10} = \sqrt{\frac{5 \pm 2\sqrt{5}}{5}}$$

Now, $\tan \theta$ is an increasing function for $0 \leq \theta < \frac{\pi}{2}$,

$$\therefore \tan \frac{3\pi}{10} > \tan \frac{\pi}{10}$$

$$\therefore \tan \frac{\pi}{10} = \sqrt{\frac{5 - 2\sqrt{5}}{5}}$$

1 $\Rightarrow$ reasoning needed

1 $\Rightarrow$ correct solution!

$$b) \quad 3x^3 + 8x^2 - 1 = 0$$

$$\alpha + \beta + \gamma = -\frac{8}{3}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 0$$

$$\alpha\beta\gamma = \frac{1}{3}$$

$$\left(\beta + \frac{1}{\gamma}\right) \left(\gamma + \frac{1}{\alpha}\right) \left(\alpha + \frac{1}{\beta}\right)$$

$$= \left(\frac{\gamma\beta + 1}{\gamma}\right) \left(\frac{\alpha\gamma + 1}{\alpha}\right) \left(\frac{\alpha\beta + 1}{\beta}\right)$$

$$= \frac{1}{\alpha\beta\gamma} (\alpha\beta\gamma^2 + \beta\gamma + \alpha\gamma + 1) (\alpha\beta + 1)$$

$$= \frac{1}{\alpha\beta\gamma} (\alpha^2\beta\gamma^2 + \alpha\beta\gamma^2 + \alpha\beta^2\gamma + \beta\gamma$$

$$+ \alpha^2\beta\gamma + \alpha\gamma + \alpha\beta + 1)$$

$$= \alpha\beta\gamma + \alpha\beta\gamma(\alpha + \beta + \gamma) + \alpha\beta + \alpha\gamma + \beta\gamma + 1$$

$$= \frac{1}{3} + \frac{1}{3} \left(-\frac{8}{3}\right) + 0 + 1$$

$$= \frac{2}{3}$$

} 1

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} 1

MATHEMATICS Extension 1 : Question 9

Suggested Solutions

Marks

Marker's Comments

$$c) i) T = 22 - Ae^{-kt}$$

$$\frac{dT}{dt} = (-k)(-Ae^{-kt})$$

$$= (-k)(22 - Ae^{-kt} - 22)$$

$$= (-k)(T - 22)$$

$\therefore T = 22 - Ae^{-kt}$ is a solution
to $\frac{dT}{dt} = (-k)(T - 22)$

$$ii) \text{ when } t=0, T=-8$$

$$-8 = 22 - Ae^{-k(0)}$$

$$A = 22 + 8$$

$$A = 30$$

$$iii) \text{ when } t=90, T=4$$

$$\therefore 4 = 22 - 30e^{-k(90)}$$

$$30e^{-90k} = 18$$

$$e^{-90k} = \frac{18}{30}$$

$$e^{-90k} = \frac{3}{5}$$

$$-90k = \ln \frac{3}{5}$$

$$k = -\ln \frac{3}{5}$$

$$90$$

$$\text{or } k = \frac{\ln \frac{5}{3}}{90}$$

$$iv) \text{ when } t=180$$

$$T = 22 - 30e^{\frac{\ln \frac{5}{3}}{90}(180)}$$

$$= 11.2$$

$$d) i) P(t) = \frac{6000}{3 + 7e^{-0.2t}}$$

$$\text{as } t \rightarrow \infty, e^{-0.2t} \rightarrow 0$$

$$\therefore P(t) \rightarrow \frac{6000}{3}$$

$$= 2000$$

$\therefore$ as $t \rightarrow \infty$, $P(t)$ approaches 2000

1
1
1
1
1
1

$\Rightarrow$ show substitution

MATHEMATICS Extension 1 : Question.....9

Suggested Solutions

Marks

Marker's Comments

ii) When $P(t) = 1000$,

$$1000 = 6000$$

$$3 + 7e^{-0.2t}$$

$$3 + 7e^{-0.2t} = 6$$

$$7e^{-0.2t} = 3$$

$$e^{-0.2t} = \frac{3}{7}$$

$$-0.2t = \ln \frac{3}{7}$$

$$t = \frac{\ln \frac{3}{7}}{-0.2}$$

$$= 4.236489\dots$$

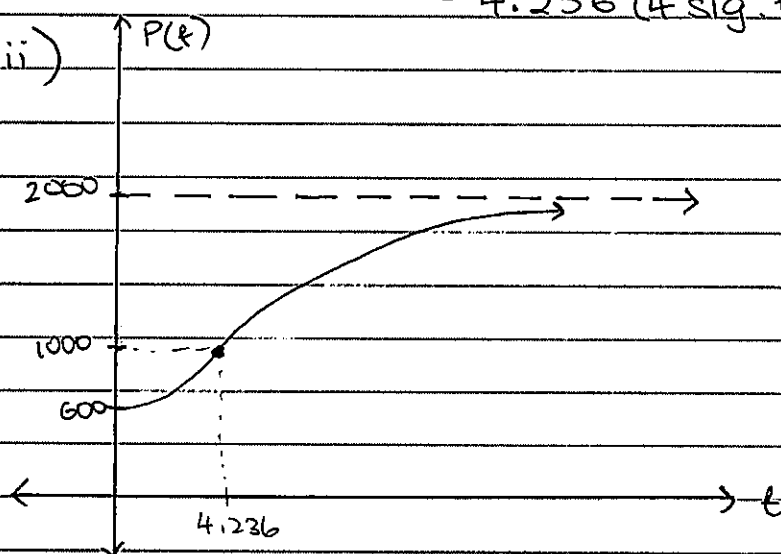
$$= 4.236 \text{ (4 sig. fig.)}$$

1

a lot of students wrote that
 $6 - 3 = 2$ X

1

iii)



1 $\Rightarrow$ asymptote and starting point

1. $\Rightarrow$ having
(4.236, 1000)
as steepest
point
ie. POI, change
in concavity
must be shown